Macroeconomics III
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Chapter 8

PRODUCTIVE EXTERNALITIES AND ENDOGENOUS GROWTH

INTRODUCTION TO ENDOGENOUS GROWTH

- In typical Western countries, income per capita has increased by factors between 6 and 8 over the last 100 years. How has this been possible?
- Technological growth: in all of our Solow models, **long-run economic growth is rooted in technological growth**. But technological growth is unexplained in these models.
- An **endogenous growth** model explains/endogenizes the long-run technological growth rate and hence the long-run growth rate of output per worker. That is, the model shows how these growth rates depend on model parameters.
- Thereby, the models imply statements on **how economic policy affects long-run growth**.

TWO TYPES OF ENDOGENOUS GROWTH MODELS

- **R&D based**: Explicitly describe the production of technological progress, i.e., contain a production function with output, $A_{t+1} - A_t$, depending on certain inputs.
- **Externality based**: No explicit production function for technological progress, but an assumption that (labour augmenting) technology in every firm, A_t , depends positively on the aggregate capital stock because of ”productive externalities”.

  This implies **increasing returns** in the aggregate production function allowing growth of GDP per worker in the long run without exogenous technological growth.
- Start with: Endogenous growth based **on productive externalities.** The new feature: $A_t = K_t^{\phi}$

A MODEL OF ENDOGENOUS GROWTH BASED ON PRODUCTIVE EXTERNALITIES

- There is one representative type of firm, which we can see in two roles:
  1. As the average producer of aggregate output
  2. As the individual, small firm that takes all aggregates as given
- When the firm decides **its** capital demand, K_t^d , it takes the **aggregate** capital stock, K_t, as given because the firm is too small to influence the economy's aggregates.
- But "at the end of the day", that is, in equilibrium, one must have $K_t^d = K_t$ because there is only this firm type.
- At **the level of the individual firm**, the production function is:

$$Y_t = \left(K_t^d\right)^{\alpha}\left(A_t L_t^d\right)^{1-\alpha}, \quad 0 < \alpha < 1,$$

where A_t is taken as given, and there is CRS to $\left(K_t^d, L_t^d\right)$, in accordance with the replication argument.

- Because of **productive externalities,** the individual firm's A_t **depends on aggregate capital**, K_t :

$$A_t = K_t^{\phi}, \ \ \phi > 0.$$

- At **the aggregate level** the production function (in equilibrium) is:

$$Y_t = K_t^{\alpha}\left(K_t^{\phi} L_t\right)^{1-\alpha} = K_t^{\alpha+\phi(1-\alpha)} L_t^{1-\alpha}.$$

  The sum of the exponents is $1+\phi\left(1-\alpha\right)$ implying that there are **IRS** if $\phi > 0$.

- **We have CRS at the individual firm level and IRS at the aggregate level.**

The micro economy is as in the Solow model:

- Competitive clearing of capital and labour markets implies "real rental rates equal to marginal products" for the inputs. But which marginal products are relevant?
- Since the optimality conditions come from **the individual** firm's factor demands, it should be the marginal products at the firm level, where A_t is taken as given:

$$r_t = \alpha \left(\frac{K_t^d}{A_t L_t^d} \right)^{\alpha - 1}, \qquad w_t = (1-\alpha) \left(\frac{K_t^d}{A_t L_t^d} \right)^{\alpha} A_t .$$

Then, inserting $K_t^d = K_t$, $L_t^d = L_t$ and $A_t = K_t^{\phi}$, etc. gives:

$$r_t = \alpha \left(\frac{K_t}{K_t^{\phi} L_t} \right)^{\alpha - 1}, \qquad w_t = (1-\alpha) \left(\frac{K_t}{K_t^{\phi} L_t} \right)^{\alpha} A_t .$$

- It follows that $r_t K_t = \alpha Y_t$ and $w_t L_t = (1-\alpha) Y_t$.
- **Our theory of the functional income distribution is as usual.**

- How can we motivate the assumption of productive externalities?
- **Empirics:** estimates of the equivalent of our model's $1+\phi(1-\alpha)$ are often larger than one, e.g., around 1.5 (a fairly large estimate, however). This gives a ϕ of around ¾, not completely ruling out $\phi = 1$.
- **Theory** (reasoning): **learning by doing**. Workers get more skilled (they are learning) as they use new capital (by doing). Workers become more productive, not only because there is more capital (the direct/internal effect), but because they learn new skills, **which they keep if they are deprived of the new capital, e.g., if they get another job** (the indirect/external effect).
- The last effect does not (in the longer run) accrue particularly to the individual firm, but to all firms.

THE COMPLETE MODEL

…consists of the equations for the factor prices plus:

$$Y_t = \left(K_t\right)^{\alpha}\left(A_t L_t\right)^{1-\alpha}, \quad 0 < \alpha < 1$$

$$A_t = K_t^{\phi},$$

$$S_t = sY_t,$$

$$K_{t+1} = S_t + \left(1-\delta\right)K_t,$$

$$L_{t+1} = \left(1+n\right)L_t.$$

Parameters: $\alpha, \phi, s, \delta, n$.

State variables: K_t and L_t .

No equation $A_{t+1} = \left(1+g\right)A_t$.

- Take another look at the aggregate production function:

$$Y_t = K_t^{\alpha}\left(K_t^{\phi} L_t\right)^{1-\alpha} = K_t^{\alpha+\phi(1-\alpha)} L_t^{1-\alpha}.$$

- If $\phi = 0$: the basic Solow model we have already studied
- Assume therefore that $\phi > 0$: **increasing returns** in the aggregate production function
- If $\phi < 1$, then $\alpha + \phi(1-\alpha) < 1$: diminishing returns to the reproducible factor, capital, alone. Leads to **”semi-endogenous growth”**.
- If $\phi = 1$, then $\alpha + \phi(1-\alpha) = 1$: constant returns to capital alone. Leads to **truly endogenous growth**.
- What about $\phi > 1$? Gives an **extreme model with accelerating growth**. Returns to capital and growth rates increase as the country gets richer: **not confirmed by the data**. Therefore we will focus on the remaining two cases ($\phi < 1$ and $\phi = 1$).

SEMI-ENDOGENOUS GROWTH ($\phi < 1$)

- Define

$$\tilde{k}_t \equiv k_t / A_t = K_t / \left(A_t L_t\right) \text{ and } \tilde{y}_t = y_t / A_t = Y_t / \left(A_t L_t\right).$$

From $Y_t = \left(K_t\right)^{\alpha} \left(A_t L_t\right)^{1-\alpha}$, we get: $\tilde{y}_t = \tilde{k}_t^{\alpha}$.

- From $A_t = K_t^{\phi}$:

$$\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t}\right)^{\phi}.$$

- Then:

$$\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{K_{t+1} / K_t}{\left(A_{t+1} / A_t\right) L_{t+1} / L_t} = \frac{K_{t+1} / K_t}{\left(K_{t+1} / K_t\right)^{\phi} L_{t+1} / L_t} = \frac{1}{1+n}\left(\frac{K_{t+1}}{K_t}\right)^{1-\phi}$$

- Inserting $K_{t+1} = S_t + \left(1-\delta\right) K_t$ gives:

$$\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{1}{1+n}\left(s\frac{Y_t}{K_t} + \left(1-\delta\right)\right)^{1-\phi} = \frac{1}{1+n}\left(s\tilde{k}_t^{\alpha-1} + \left(1-\delta\right)\right)^{1-\phi}.$$

- Rearranging gives the transition equation:

$$\tilde{k}_{t+1} = \frac{1}{1+n}\tilde{k}_t\left(s\tilde{k}_t^{\alpha-1} + (1-\delta)\right)^{1-\phi} = \frac{1}{1+n}\left(s\left(\tilde{k}_t\right)^{\frac{\alpha+\phi-\alpha\phi}{1-\phi}} + (1-\delta)\left(\tilde{k}_t\right)^{\frac{1}{1-\phi}}\right)^{1-\phi}$$

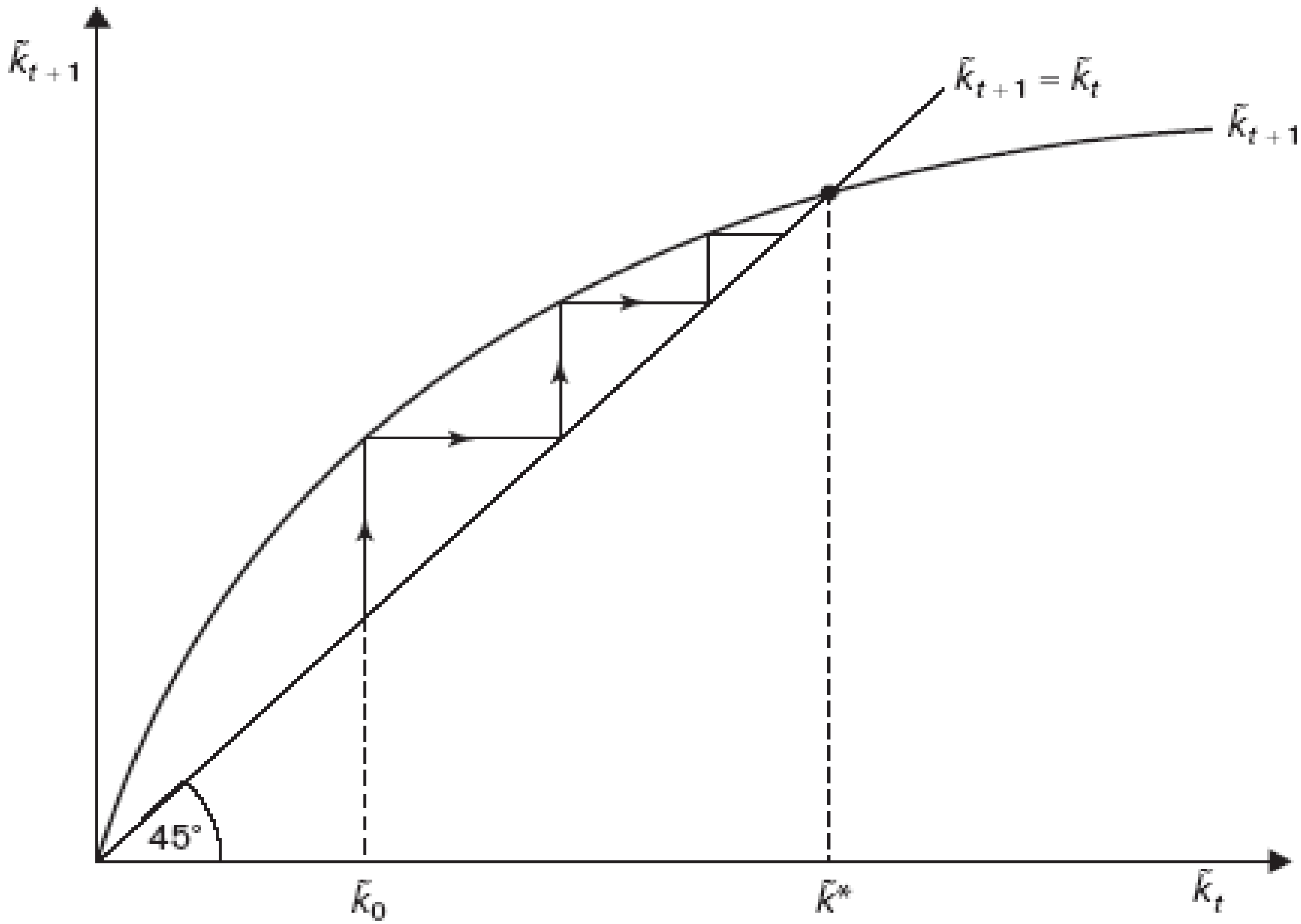
Note: the exponents in the last expression are all positive. The transition equation has the following properties:

1. It passes through (0,0).
2. It is everywhere increasing.
3. There is a unique positive intersection, $\tilde{k}^*$, with the 45° line. Inserting $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}$ gives:

$$1+n = \left(s\tilde{k}^{\alpha-1} + (1-\delta)\right)^{1-\phi} \Leftrightarrow (1+n)^{1/(1-\phi)} - (1-\delta) = s\tilde{k}^{\alpha-1} \Leftrightarrow$$

$$\tilde{k} = \left(\frac{s}{(1+n)^{1/(1+\phi)} - (1-\delta)}\right)^{1/(1-\alpha)} \equiv \tilde{k}^* > 0 \quad (\text{we assume that } n+\delta > 0)$$

4. The slope in $\tilde{k}^*$ is less than one (differentiate, etc. and use $n+\delta>0$).



- $\tilde{k}_t$ converges to $\tilde{k}^*$ implying that $\tilde{y}_t$ converges to:

$$\tilde{y}^* = \left(\tilde{k}^*\right)^{\alpha} = \left(\frac{s}{(1+n)^{1/(1+\phi)} - (1-\delta)}\right)^{1/(1-\alpha)}.$$

This defines steady state.

GROWTH IN STEADY STATE

- When $\tilde{k}_t \equiv k_t / A_t$ and $\tilde{y}_t = y_t / A_t$ have converged to the constant steady state values $\tilde{k}^*$ and $\tilde{y}^*$, respectively, k_t and y_t must grow at the same rate as A_t.
- The growth rate of A_t is endogenous! We can easily find its value in steady state:
- In steady state $\tilde{k}_{t+1} / \tilde{k}_t = 1$, from which:

$$\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{1}{1+n}\left(\frac{K_{t+1}}{K_t}\right)^{1-\phi} = 1 \Leftrightarrow \frac{K_{t+1}}{K_t} = (1+n)^{1/(1-\phi)} \Rightarrow$$

$$\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t}\right)^{\phi} = (1+n)^{\phi/(1-\phi)} \Leftrightarrow$$

$$\frac{A_{t+1} - A_t}{A_t} = (1+n)^{\phi/(1-\phi)} - 1 \equiv g_{se}.$$

- Thus, our model implies convergence to a steady state with a common constant growth rate of k_t, y_t and A_t:

$$g_{se} = (1+n)^{\phi/(1-\phi)} - 1.$$

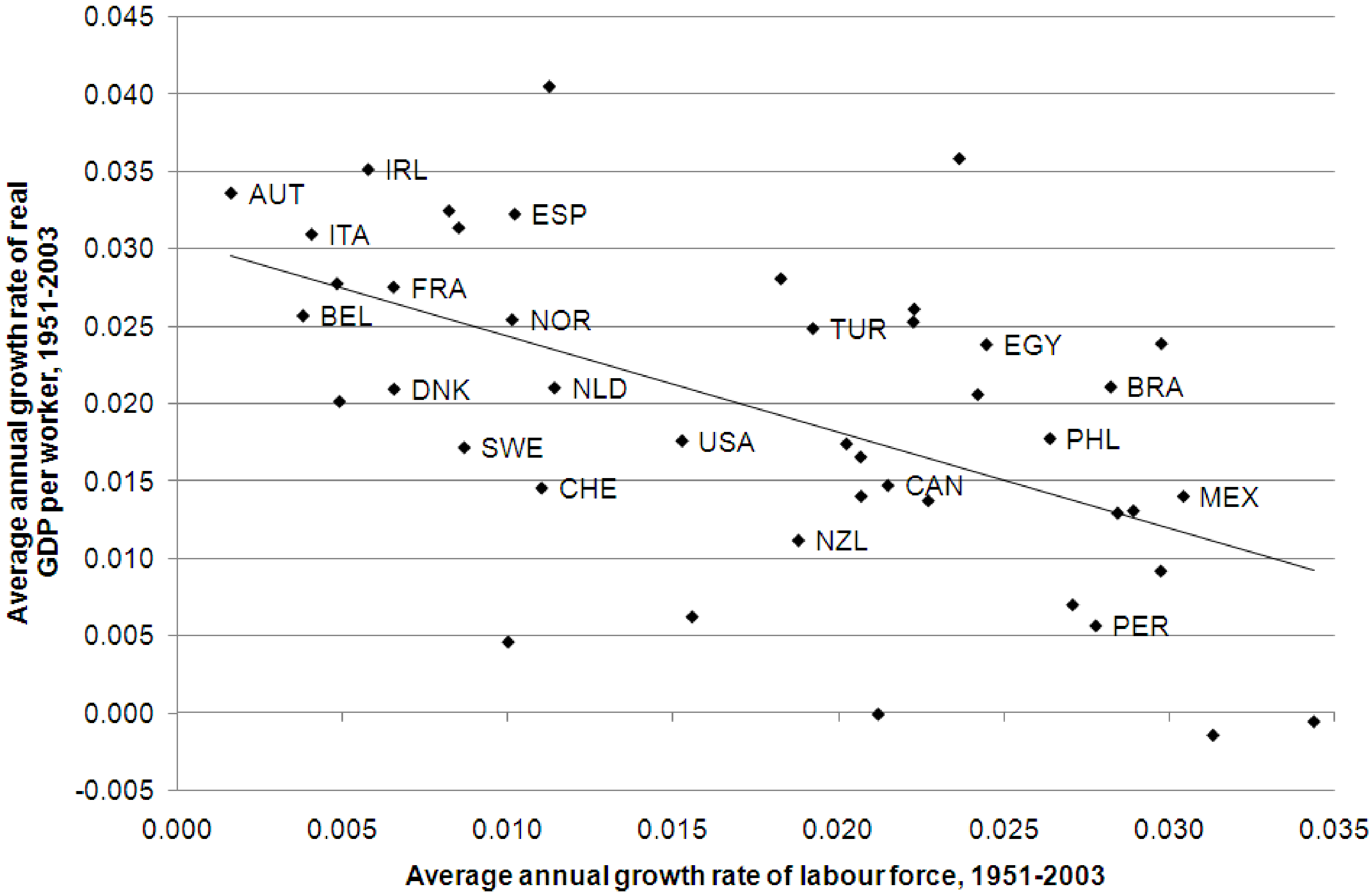
- In fact, there is **balanced growth** in steady state
- Percapita GDP and consumption can grow forever.
- We have a steady state with **persistent growth:** positive growth in y_t without exogenous technical progress, and the growth rate depends on model parameters.
- But we only have $g_{se} > 0$ if $n > 0$: labour force growth is required for percapita economic growth.
- Our steady state is therefore one of **semi-endogenous growth:** there is only economic growth if the labour force grows!
- Intuition: to utilize increasing returns, scale increases are required.

- The most obvious implication for structural policy is: in order to promote long-run economic growth, promote population growth!

- Reason for being cautious with such a policy: the empirical evidence does not support its prediction.

EMPIRICS FOR SEMI-ENDOGENOUS GROWTH

- Plotting g^i against n^i, 1951-2003, across 44 countries:



In contradiction to semi-endogenous growth, there is a clear negative correlation between g^i and n^i across countries, but:

- Direction of causality?
- Is our model a country by country model or is it for the World?
- The figure is in accordance with the model's **transitory growth** if convergence is slow: perhaps we should not reject the model, but only the idea that the countries are **in** steady state?
- Perhaps we should look at even longer periods and not look across countries.

	Average annual population growth rate (%)		Average annual growth rate of GDP per capita (%)	
	1870–1930	1930–1990	1870–1930	1930–1990
Australia	2.3	1.6	0.4	2.0
Austria	0.7	0.2	1.1	2.6
Belgium	0.8	0.4	1.0	2.1
Canada	1.7	1.6	1.7	2.4
Denmark	1.0	0.6	1.6	2.1
Finland	1.1	0.6	1.4	3.1
France	0.1	0.5	1.5	2.3
Germany	1.0	0.7	1.2	2.5
Italy	0.6	0.6	1.1	2.9
Japan	1.0	1.1	1.5	3.9
Netherlands	1.3	1.1	1.2	1.8
New Zealand	2.7	1.4	1.2	1.6
Norway	0.8	0.7	1.6	2.7
Sweden	0.6	0.6	1.4	2.5
Switzerland	0.7	0.8	1.7	2.1
UK	0.7	0.4	0.8	1.9
USA	1.9	1.2	1.5	2.1

- Country by country tendency: population growth decreases and economic growth increases from the first to the second subperiod. This speaks against semi-endogenous growth.
- However, in an even wider, global perspective, one can counterargue that the last 200 years form **both** the period in which the World has seen non-negligible average annual growth rates in income per capita **and** the period in which there has been non-negligible average annual population growth rates.
- If convergence is **very** slow, lower population growth in combination with higher economic growth can still be in accordance with the transitory growth of the model. But if convergence is that slow the steady state itself is not so interesting; rather, the process of convergence to steady state is essential.

ENDOGENOUS GROWTH ($\phi = 1$): THE AK MODEL

- In the model just considered ($\phi < 1$) convergence becomes very slow as $\phi \to 1$ (exercises), and slow convergence is what we want to study now. Assuming that $\phi = 1$, we approximate the case of a large ϕ just below one. We also assume $n = 0$, and thus $L_t = L$. Otherwise our model becomes extreme.
- The complete model boils down to two equations, derived from the aggregate production function and the capital accumulation equation, respectively:

$$Y_t = K_t^{\alpha}\left(A_t L\right)^{1-\alpha} = K_t^{\alpha}\left(K_t^{\phi} L\right)^{1-\alpha} = K_t^{\alpha+\phi(1-\alpha)} L^{1-\alpha} = K_t L^{1-\alpha},$$

$$K_{t+1} = sY_t + \left(1-\delta\right)K_t.$$

- The AK model:

$$Y_t = AK_t, \quad A \equiv L^{1-\alpha},$$

$$K_{t+1} = sY_t + (1-\delta)K_t.$$

- Dividing on both sides of both equations by L gives $y_t = Ak_t$ and $k_{t+1} = sy_t + (1-\delta)k_t$ which combine to:

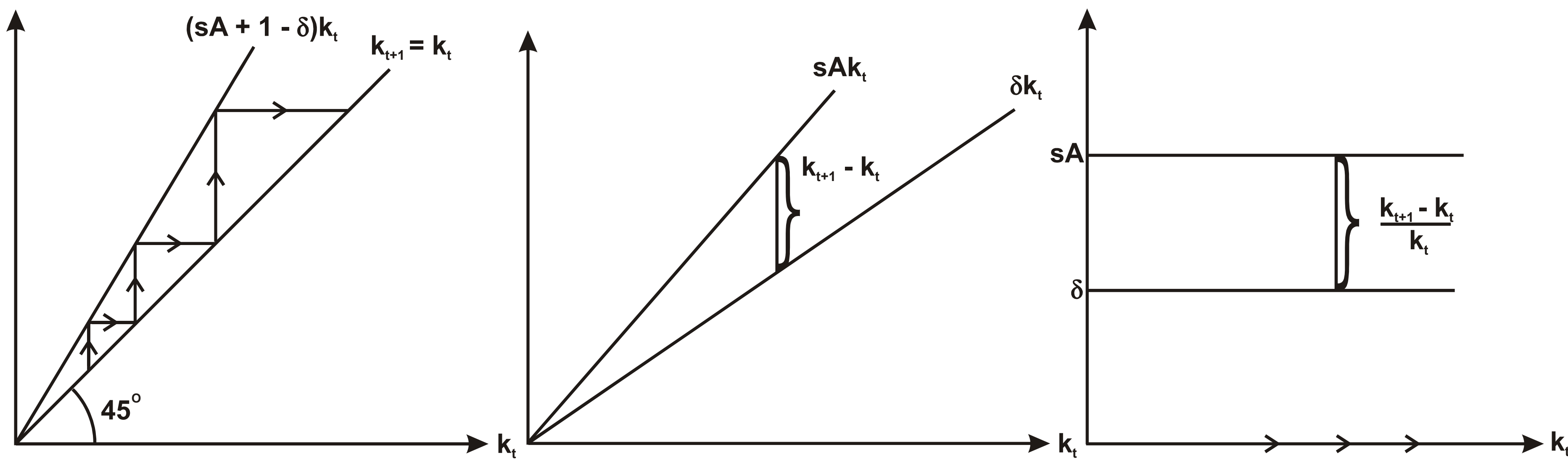
$$k_{t+1} = sAk_t + (1-\delta)k_t = (sA + 1 - \delta)k_t \Leftrightarrow$$

$$k_{t+1} - k_t = (sA - \delta)k_t \Leftrightarrow$$

$$\frac{k_{t+1} - k_t}{k_t} = sA - \delta \equiv g_e.$$

We assume that $sA > \delta$ ensuring that $g_e > 0$.

- Since $y_t = Ak_t$ and $A_t = K_t = k_t L$, all of k_t, y_t and A_t grow at the rate g_e: there is balanced growth.



- We do not have convergence to a steady state: k_t grows at rate g_e all the time.

- We have one common, constant growth rate for k_t, y_t and A_t:

$$g_e = sA - \delta$$

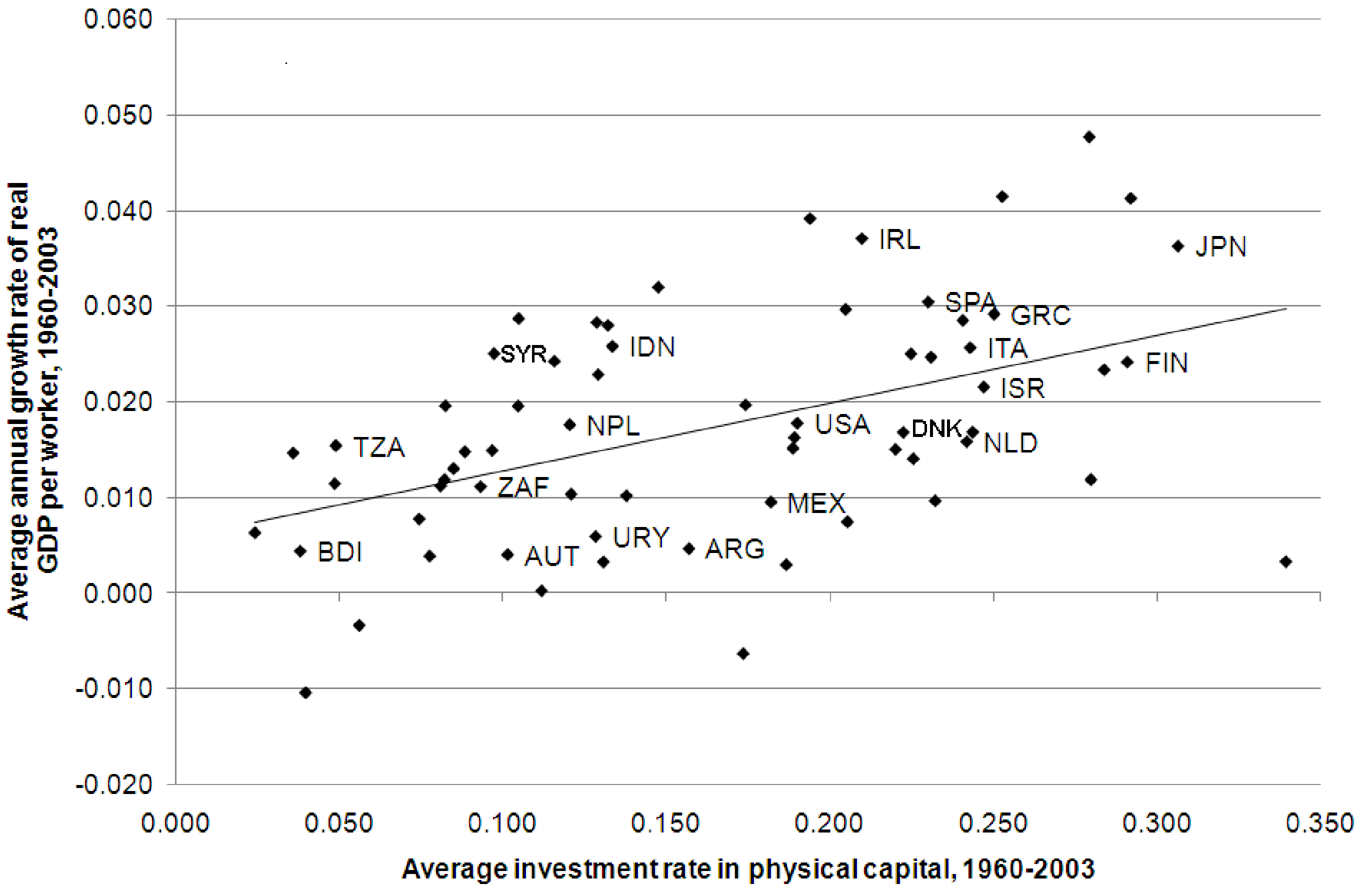
- **Truly endogenous growth:** positive growth in y_t without exogenous technological growth or population growth; the growth rate is given by model parameters, among these s.
- **Structural policies:** higher s and lower δ give **everlasting** higher growth in GDP and consumption per capita.

Critique:

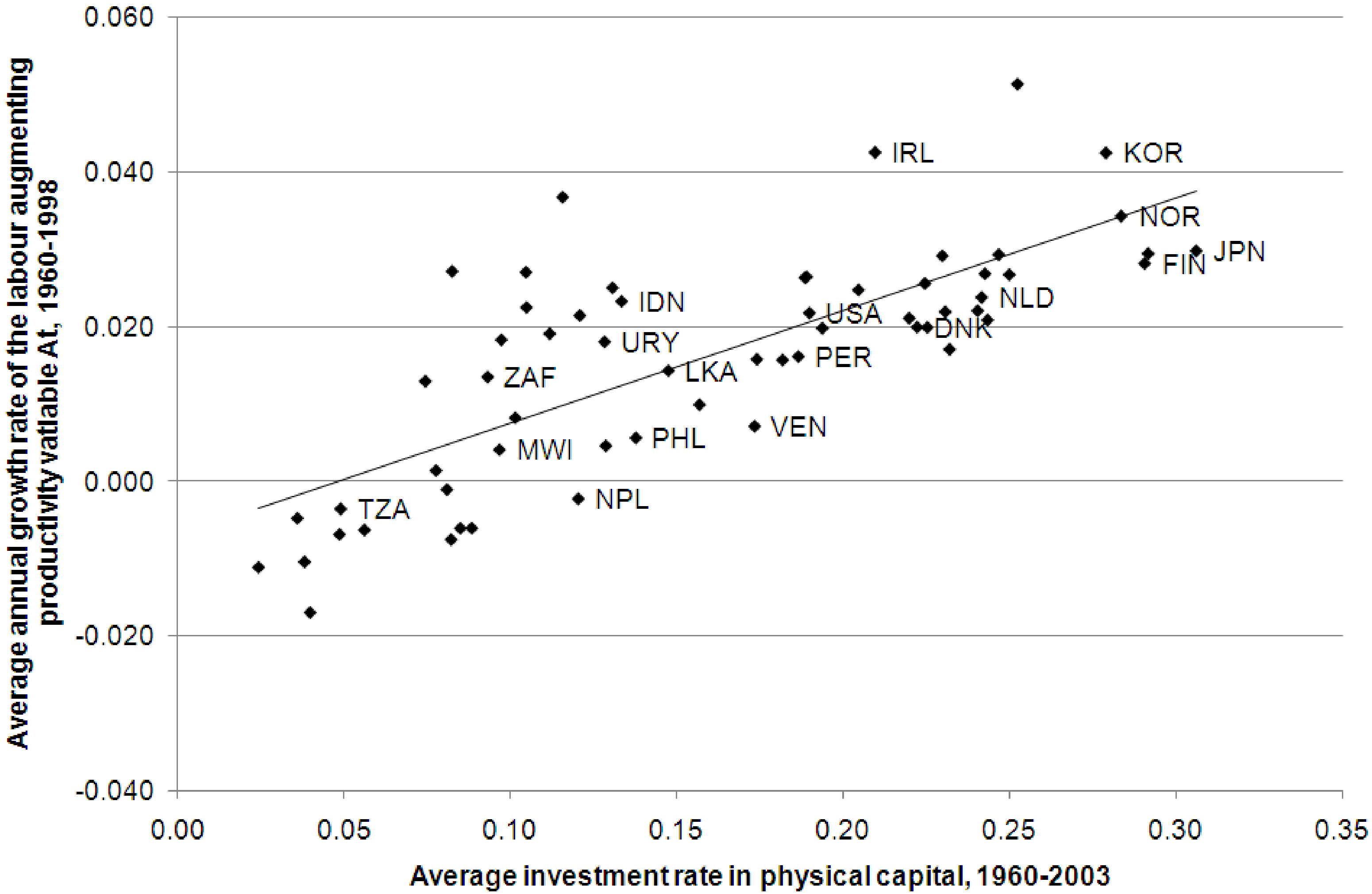
- $\phi = 1$ is a **knife-edge** case (not a valid objection, $\phi = 1$ is an approximation of ϕ less than, but close to one).
- **Scale effect:** remember that $A = L^{1-\alpha}$. A larger constant population results in higher growth and an increasing population gives accelerating growth. **Not realistic!**

Empirics for endogenous growth

- The most important prediction: larger rate of investment, s^i, increases the growth rate, g_e. Plotting g^i against s^i, 1960-2003, across 65 countries gives:



- This is in nice accordance with the **endogenous** growth model. But it could also be explained as the **transitory** growth of an **exogenous** growth model.
- The latter is **not** the case, however, if there is a positive correlation between s and the growth rate of the technology variable, A_t (this rate is exogenous in exogenous growth models, but equal to $g_e = sA - \delta$ in this model).
- Plotting the growth rate of A_t as determined by growth accounting, 1960-1998, against s^i, 1960-2003, across 65 countries gives:



This is probably the most important single argument in favour of endogenous growth models.

CONCLUSIONS

1. Productive externalities, or learning-by-doing spillovers can motivate increasing returns to capital and labour at the aggregate level at the same time as constant returns at the firm level.
2. This can result in a well behaved growth model with long-run growth in income per worker without this being generated by exogenously assumed technological progress.
3. Sufficiently weak spillovers lead to semi-endogenous growth, where the long-run economic growth rate is mainly determined by, and depends positively on, the growth rate of the labour force. Policy implication: promote labour force (or population) growth! The empirics suggest caution with such a recommendation.

4. Strong spill-overs lead to truly endogenous growth, where the long-run economic growth rate depends positively on the investment rate. This feature is empirically plausible! Policy implication: enhancing savings and investment is even more favourable than according to exogenous growth models. **But:** truly endogenous growth models exhibit an implausible scale effect!